ApexPlanet Internship

Task 1 – Foundations & Environment Setup

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Domain: Cybersecurity & Ethical Hacking
Duration: Days 1–12

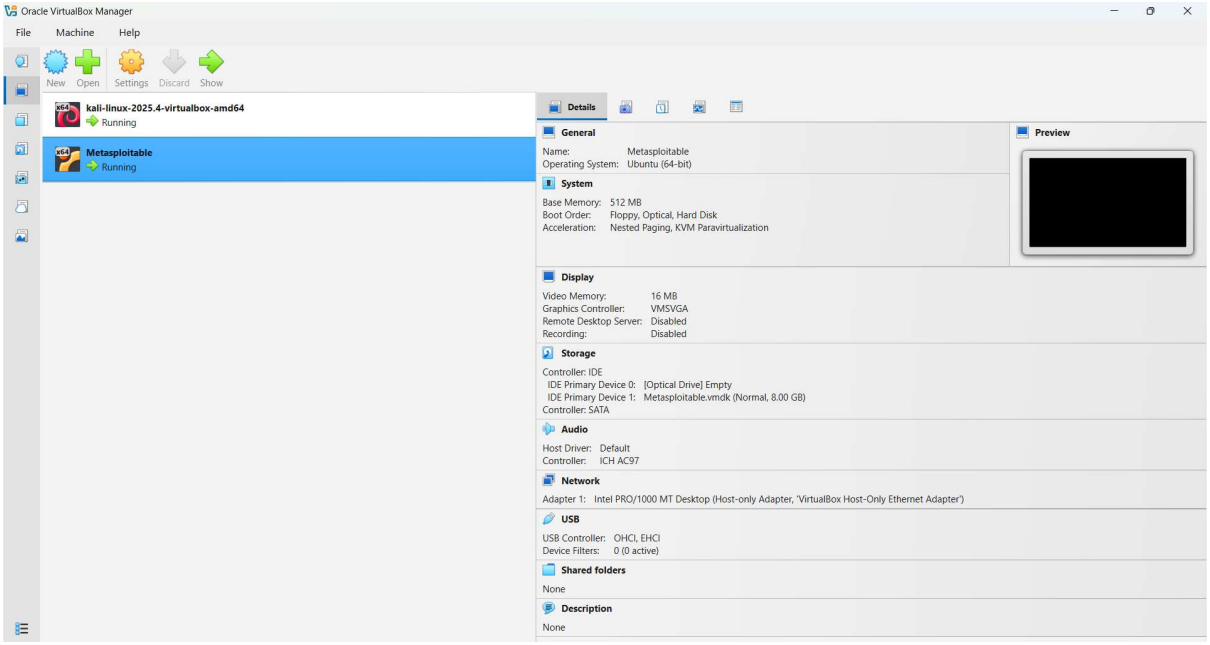
1.Introduction

This task focuses on building strong fundamentals in cybersecurity, networking, cryptography, and setting up a professional penetration testing lab environment using Kali Linux and Metasploitable.

2.Lab Environment Setup

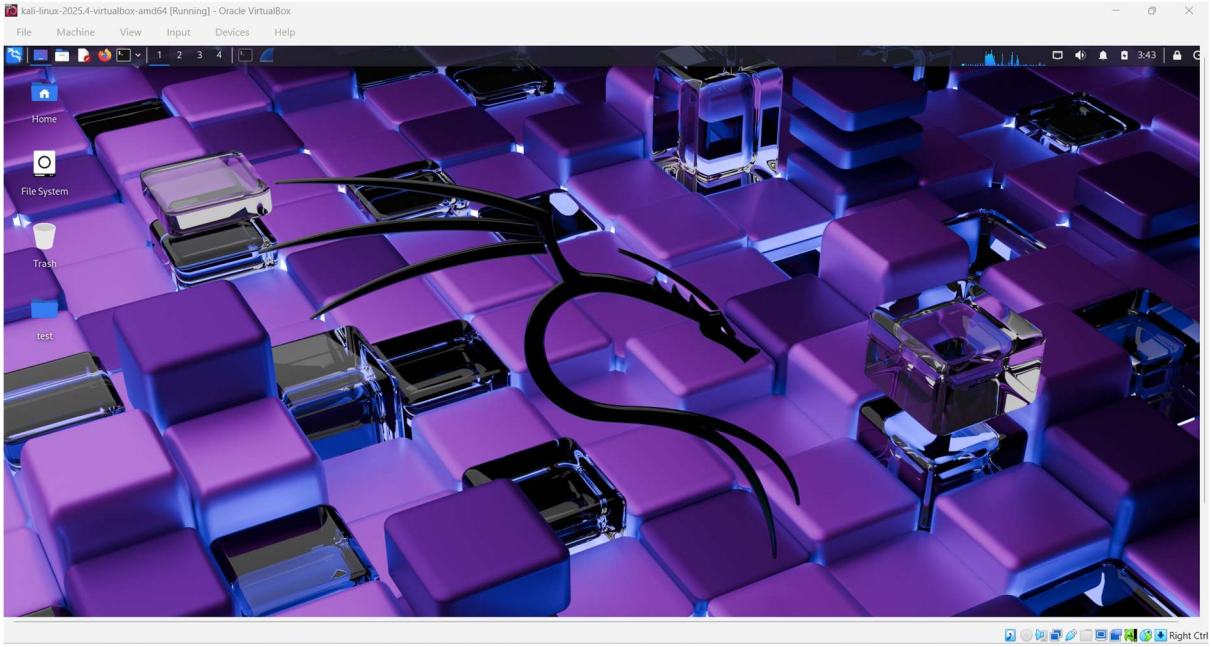
VirtualBox Installation

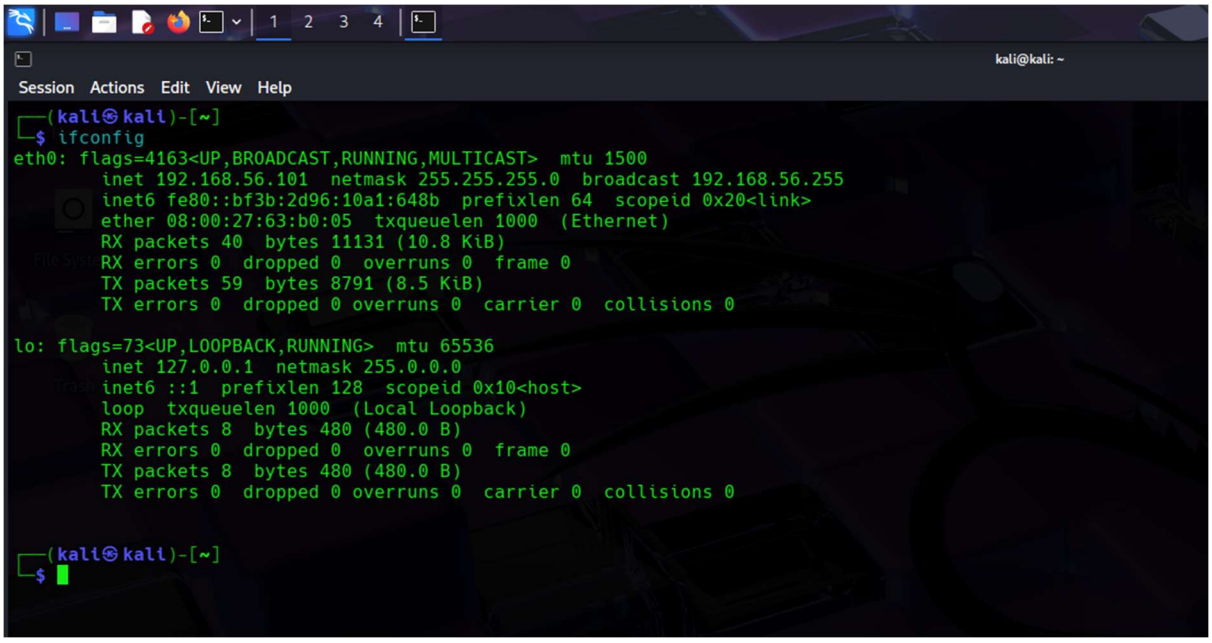
- Installed Oracle VirtualBox.
- Configured 4GB RAM.
- Created virtual machines



3.Kali Linux Setup

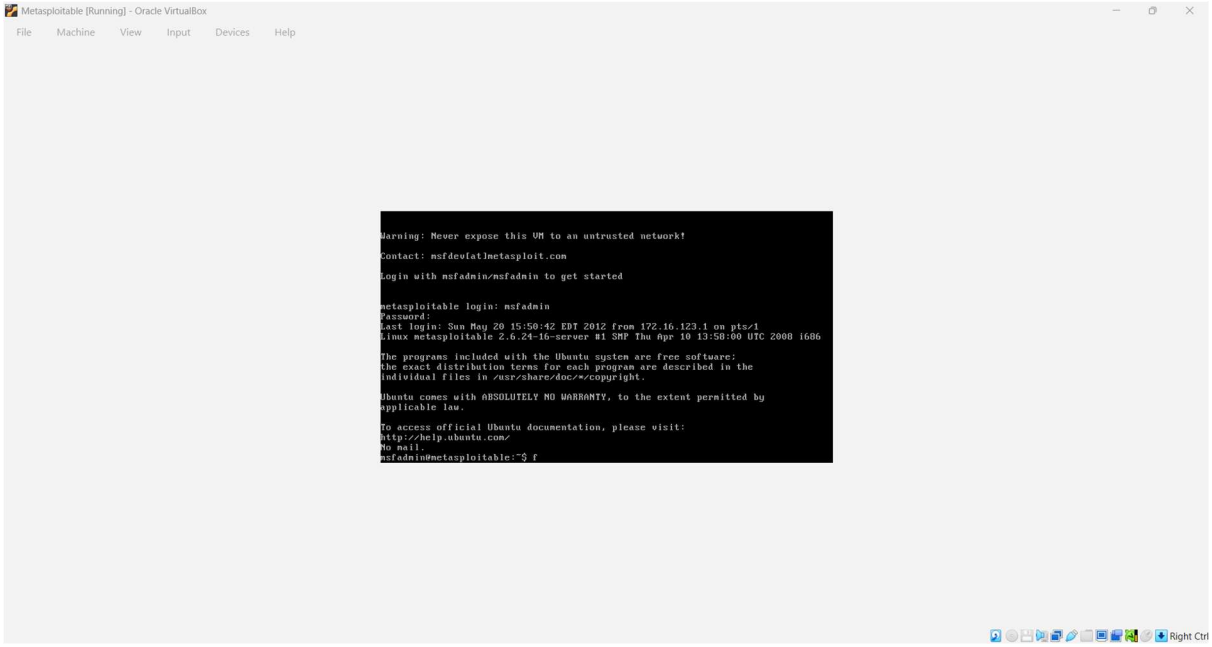
- Installed Kali Linux as attacker machine.
- Assigned Host-Only Adapter.
- Verified IP using ifconfig.





4.Metasploitable Setup

- Imported Metasploitable2.
- Login: msfadmin
- Configured Host-only network.

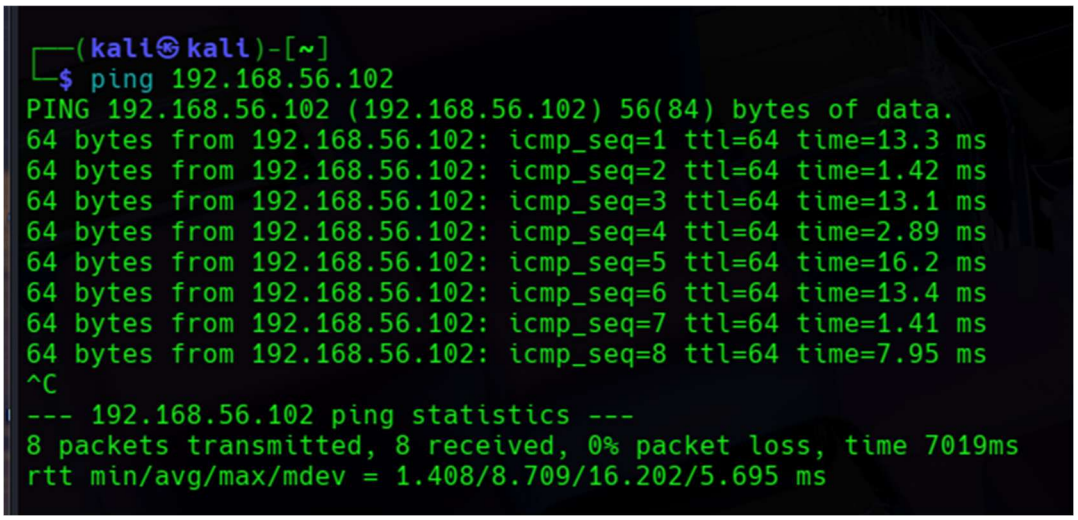


5.Network Configuration & Testing

- Both machines configured with Host-Only adapter.
- Verified connectivity using ping.

Command:

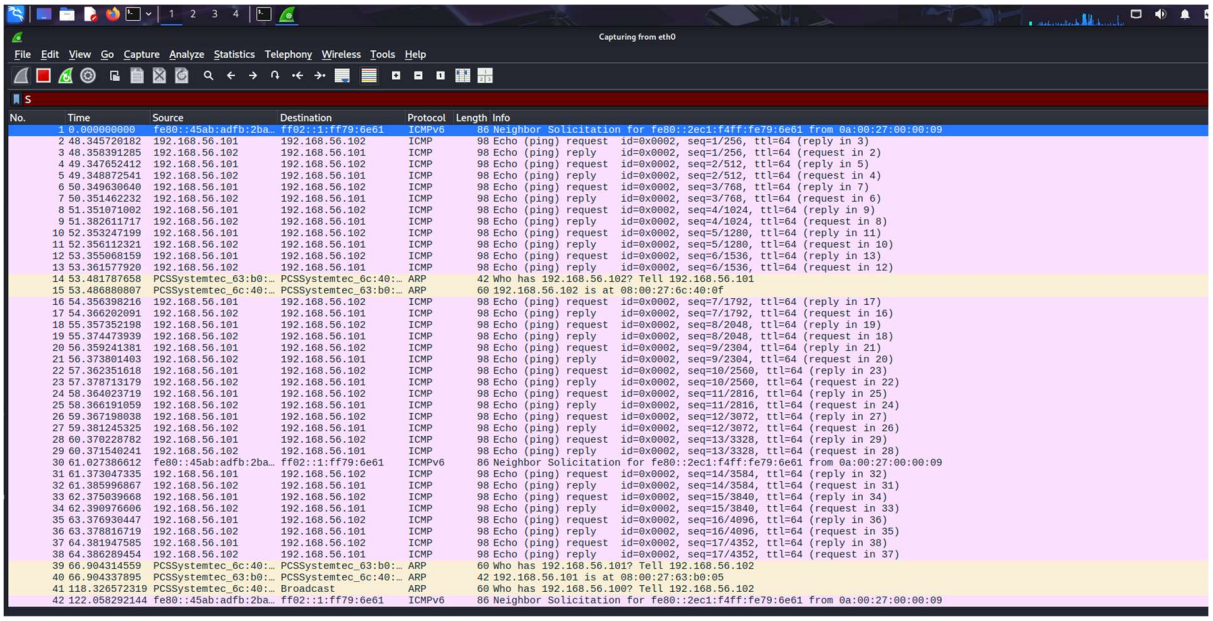
ping <metasploitable_ip>



```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=13.3 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.42 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=13.1 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.89 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=16.2 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=13.4 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.41 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=7.95 ms
^C
--- 192.168.56.102 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7019ms
rtt min/avg/max/mdev = 1.408/8.709/16.202/5.695 ms
```

6.Wireshark Packet Capture

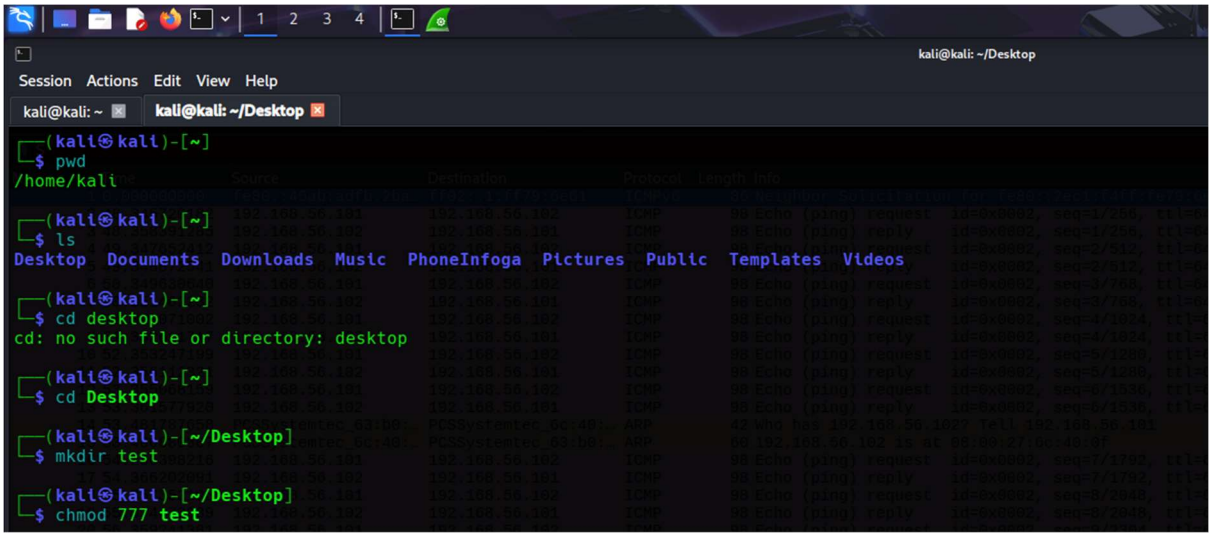
- Started capture on eth0.
- Generated traffic using ping.
- Observed ICMP packets.



7.Linux Fundamentals Practiced

Commands executed:

pwd, ls, cd, chmod, apt, update, ifconfig, netstat

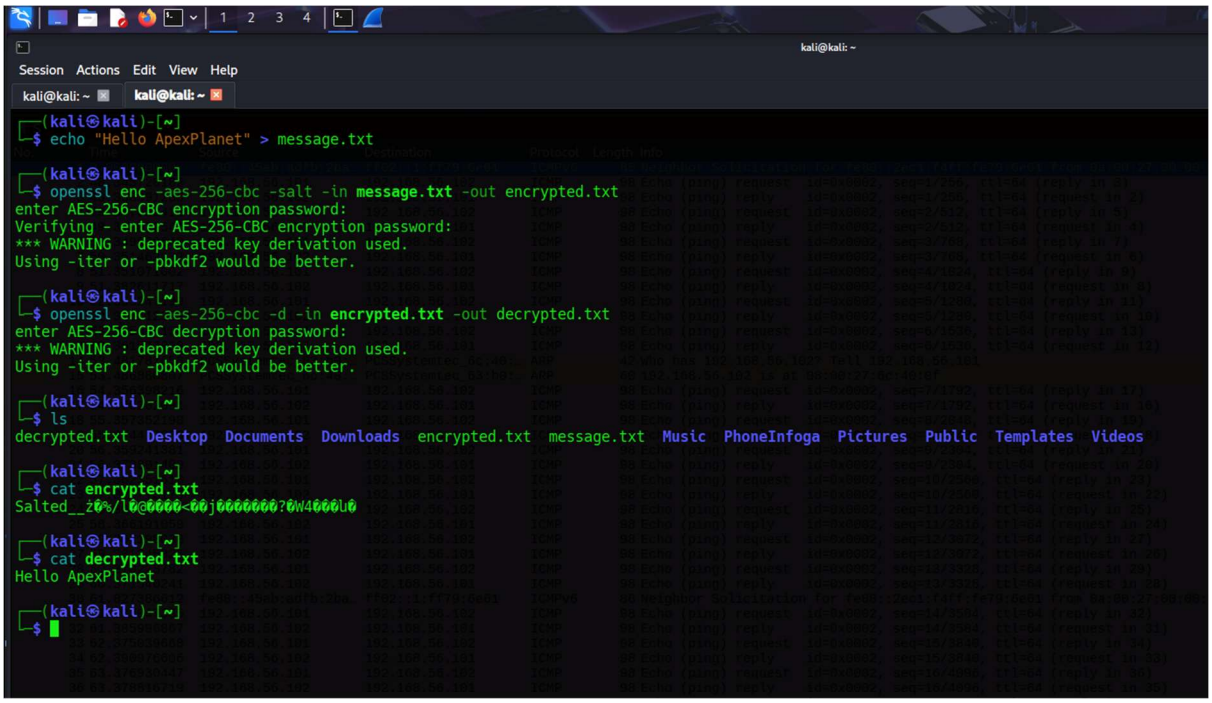


8.Cryptography Demo (OpenSSL)

Encrypted and decrypted a file using AES-256-CBC.

Commands:

openssl enc -aes-256-cbc -salt -in message.txt -out encrypted.txt
openssl enc -aes-256-cbc -d -in encrypted.txt -out decrypted.txt



9.Tools Familiarized

- Wireshark
- Nmap
- Netcat
- Burp Suite

10.Conclusion

This task helped me understand core cybersecurity concepts and build a professional penetration testing lab environment. The setup will be used for further advanced tasks in the internship.